ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Methods of Separation in Everyday Life** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** magnet	Removes iron.
2	**(d)** unsaturated	More solute can dissolve.
3	**(d)** to avoid spurting	Splashing.
4	**(a)** residue	It stays on paper.
5	**(b)** crystallises out	Solubility falls.
6	**(c)** to hold the paper well	Good contact.
7	**(b)** salt solution	Salt dissolved.
8	**(a)** solution became saturated	Extra salt remains.
9	**(c)** pure water is collected	Distillation.
10	**(a)** evaporation	Heat.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sand does not dissolve (1); Sedimentation and filtration are suitable (1).	1+1
12	Both dissolve (1); They pass through together (1).	1+1
13	It remains behind (1); The clear liquid is the filtrate (1).	1+1
14	Use a magnet to remove iron (1); Sugar remains (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Handpick stones (1); Use magnet for nails (1); Sieve out sand (1).	1+1+1
16	Large surface area (1); Sun heats quickly (1); Faster evaporation (1).	1+1+1
17	Chalk particles are tiny (1); They pass through the holes along with the water (1); Sedimentation or filtration is better (1).	1+1+1
	OR: Heat a few drops on a glass slide (1); Water evaporates (1); White solid remains (1).	1+1+1
18	Pure substance has one kind of matter (1); Mixture has more (1); Examples: gold vs sea water (1).	1+1+1
	OR: If solid is insoluble: filtration (1); If dissolved: evaporation (1); Check solubility (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Magnet removes iron (1); Add water (1); Sawdust floats and is skimmed (1); Filter sand (1); Evaporate to get sugar (1).	1+1+1+1+1
	OR: Winnowing and sieving (1); Straining (1); Evaporation (1); Churning (1); Basic ideas same (1).	1+1+1+1+1
20	Take two glasses with equal water (1); One cold, one warm (1); Add salt spoon by spoon (1); Count spoons dissolved (1); Warm water dissolves more (1).	1+1+1+1+1
	OR: Heat the solution (1); Steam to a cool vessel (1); Water collected (1); Solid remains (1); Distillation (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Methods of Separation in Everyday Life** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** evaporation of salt solution	Salt can be recovered.
2	**(a)** churning	Spins.
3	**(b)** decantation after two layers form	Immiscible liquids.
4	**(b)** sand and water	Insoluble solid.
5	**(b)** winnowing and handpicking	Combine.
6	**(a)** sedimentation	Gravity.
7	**(c)** sugar and water	Sugar dissolves.
8	**(c)** evaporate the water	Crystallisation.
9	**(a)** filtration	Strainer.
10	**(d)** magnet then dissolving and evaporating	Combination.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Size, weight, solubility differ (1); A suitable method uses one such property (1).	1+1
12	Use a sieve whose holes let sand pass but not the seeds (1); Seeds are larger than sand grains (1).	1+1
13	Add water; sugar dissolves (1); Filter and evaporate the filtrate (1).	1+1
14	They mix completely (1); They form a single layer, so nothing can be poured off (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	All have holes (1); Hole size differs (1); Used for different sizes (1).	1+1+1
16	Stones are heavier and settle (1); Decant dirty water (1); Rice is clean (1).	1+1+1
17	Threshing frees grain from stalk (1); Winnowing removes husk (1); Both done in succession (1).	1+1+1
	OR: Oil floats as a separate layer (1); Salt dissolves uniformly (1); Distillation or evaporation used for salt (1).	1+1+1
18	Layers of gravel, sand (1); Pour water (1); Dirt trapped (1).	1+1+1
	OR: Use a magnet (1); Nails stick to it (1); Beads remain (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sea water let into pans (1); Sun evaporates water (1); Salt crystals appear (1); Collected and dried (1); Purified for use (1).	1+1+1+1+1
	OR: Solution: salt in water (1); The dissolved salt cannot be seen (1); Mixture of solids: sand and sugar (1); Parts of a solid mixture can be seen (1); Parts of a solution do not settle (1).	1+1+1+1+1
20	Settle (1); Alum (1); Cloth filter (1); Boil (1); Store safely (1).	1+1+1+1+1
	OR: Stir salt in water (1); Disappears (1); Stir sand (1); Settles (1); Salt solution clear, sand residue (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Methods of Separation in Everyday Life** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** carries away dirt	Removes dirt.
2	**(a)** sieving	Different sizes.
3	**(a)** winnowing	Light and heavy.
4	**(b)** particles settle	Sedimentation.
5	**(d)** settle faster	Alum coagulates particles.
6	**(b)** dissolving sugar and filtering	Solubility differs.
7	**(d)** sieving	Size.
8	**(b)** sedimentation, decantation, filtration, boiling	Standard.
9	**(d)** evaporation	Evaporation is for dissolved solids in liquids.
10	**(a)** to remove iron pieces	Safety.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Holds the filter paper (1); Directs the filtrate (1).	1+1
12	Particles clump together (1); They settle faster (1).	1+1
13	Wind blows away light husk (1); Heavy grain falls down (1).	1+1
14	Evaporate and condense the vapour (1); Collect the droplets (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Heat a little sea water in a dish (1); The water evaporates (1); White salt is left behind (1).	1+1+1
16	Boil the water (1); Steam goes to a cool tube (1); Condenses (1).	1+1+1
17	Faster dissolving (1); Brings fresh water to the solute (1); Solution forms sooner (1).	1+1+1
	OR: Tea: strainer (1); Flour: sieve (1); Rice and stones: handpicking (1).	1+1+1
18	Use a strainer (1); Water passes (1); Rice remains (1).	1+1+1
	OR: Holds filter paper (1); Guides the liquid (1); Prevents spilling (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Heat flask (1); Steam goes through tube (1); Cooled in water (1); Condensed (1); Collect pure water (1).	1+1+1+1+1
	OR: Stones harm teeth (1); Iron bits dangerous (1); Sieves and magnets used (1); Hygiene (1); Quality (1).	1+1+1+1+1
20	Winnowing: weight (1); Magnet: magnetic property (1); Sedimentation: density (1); Evaporation: water turns to vapour but the dissolved solid does not (1); Sieving: size (1).	1+1+1+1+1
	OR: Filter juice (1); Boil to evaporate water (1); Cool (1); Crystals form (1); Separate and dry (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.